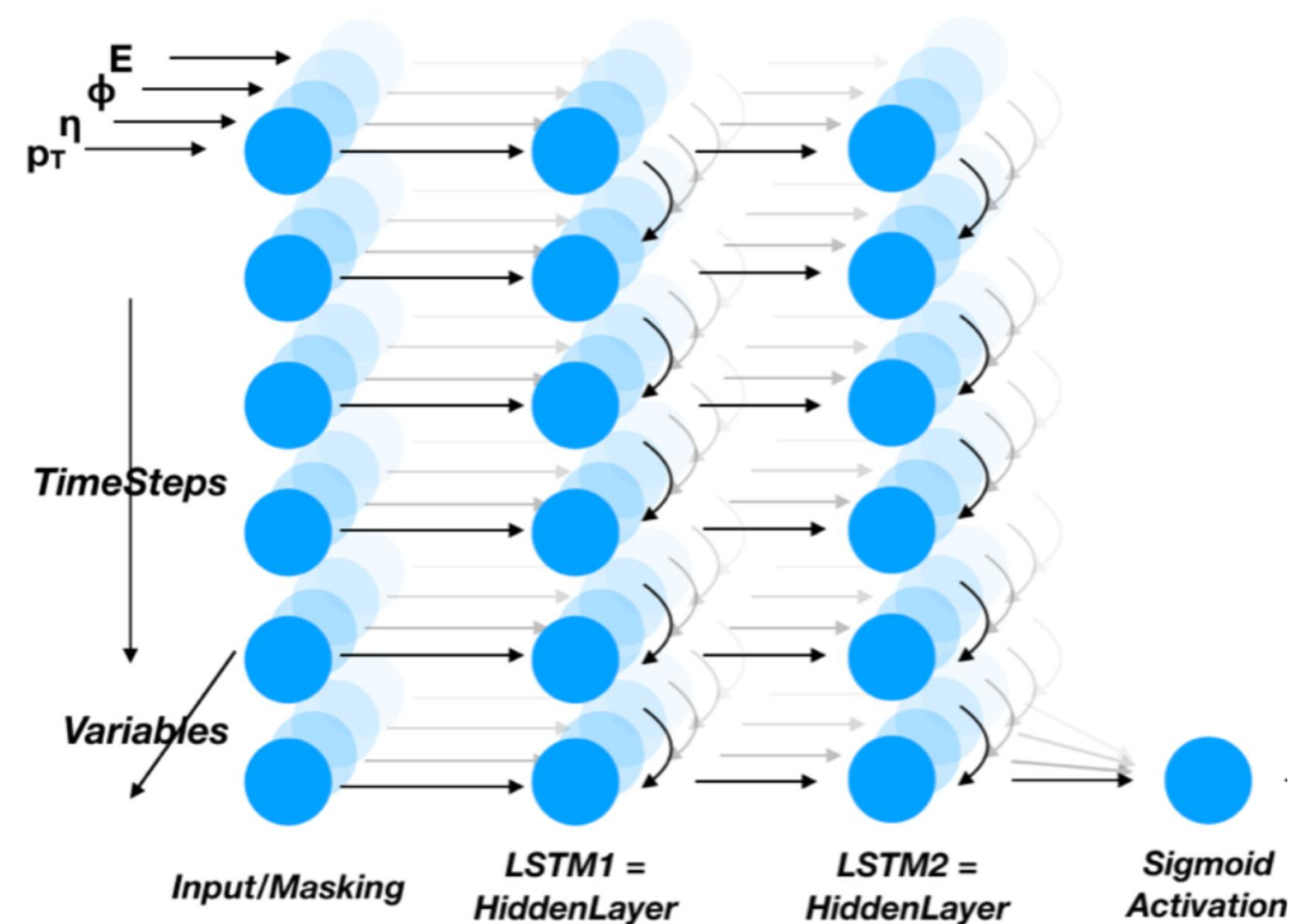
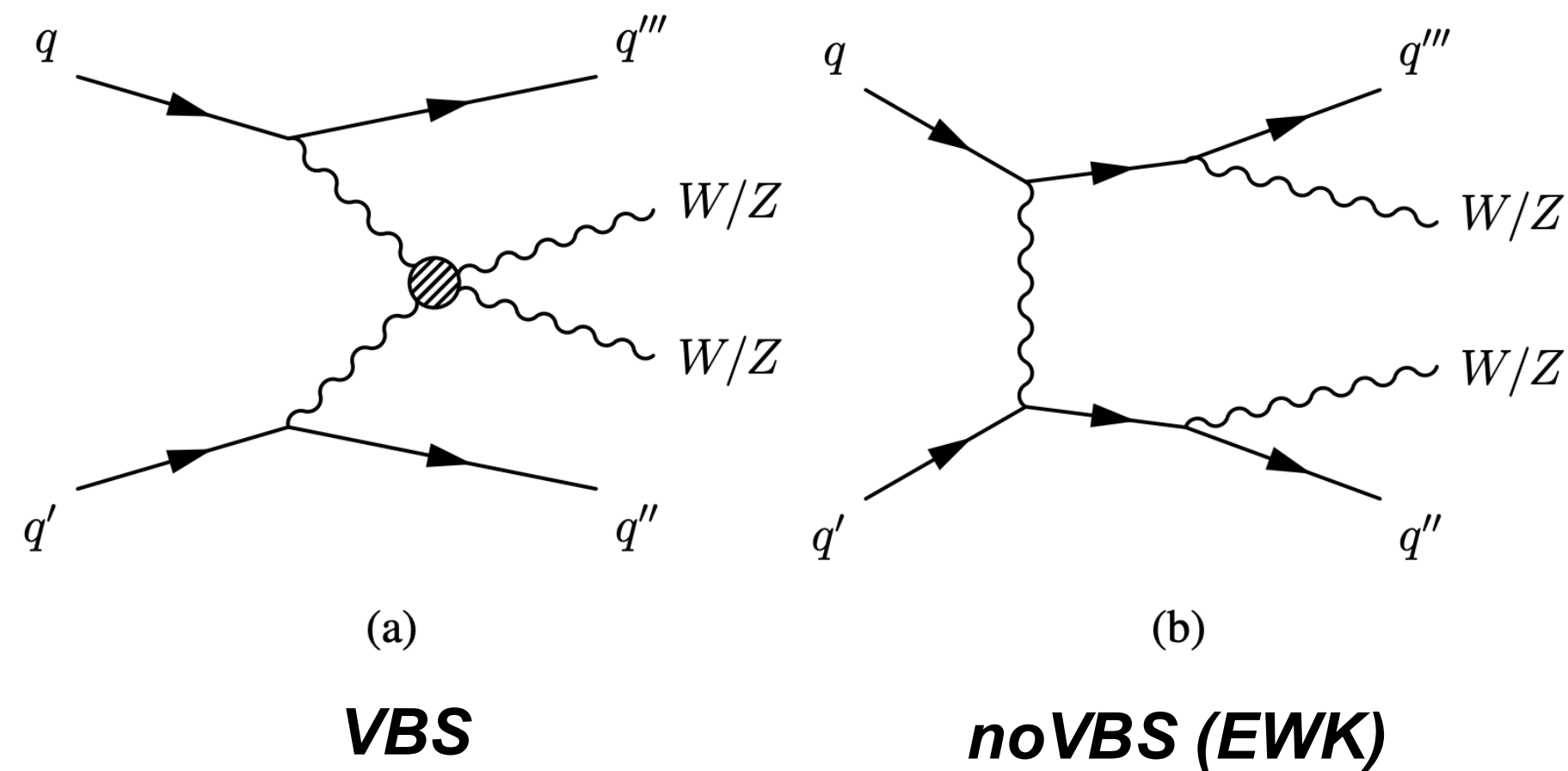


VBF-RNN

ARCHITECTURE & TRAINING



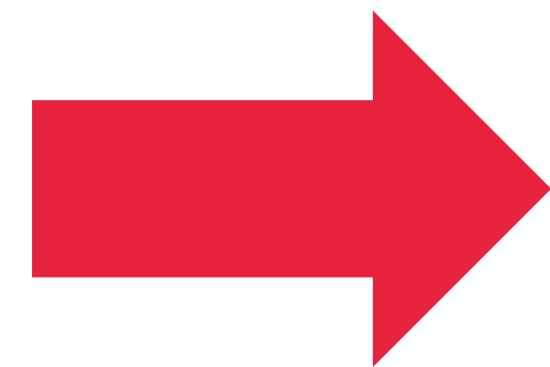
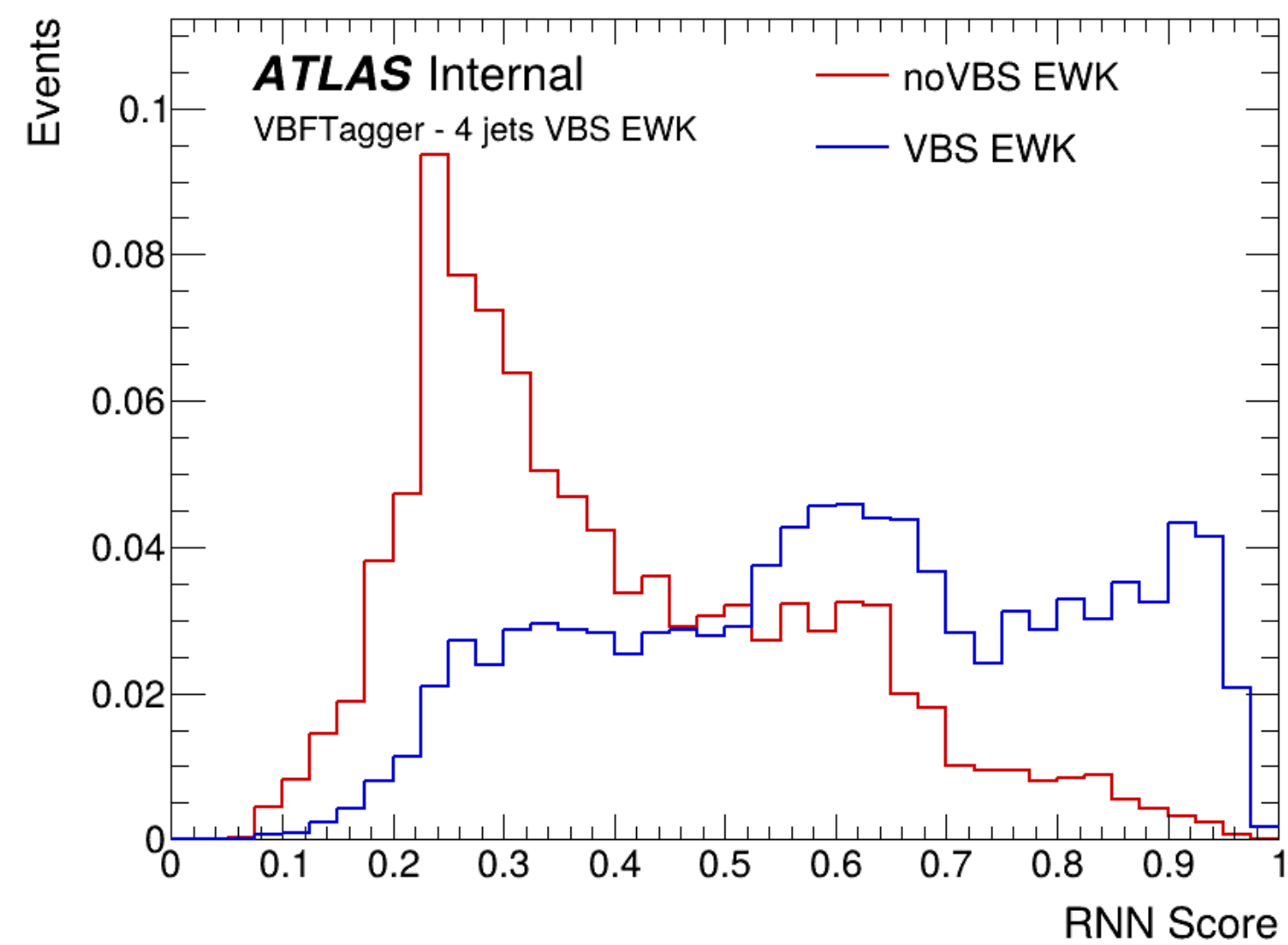
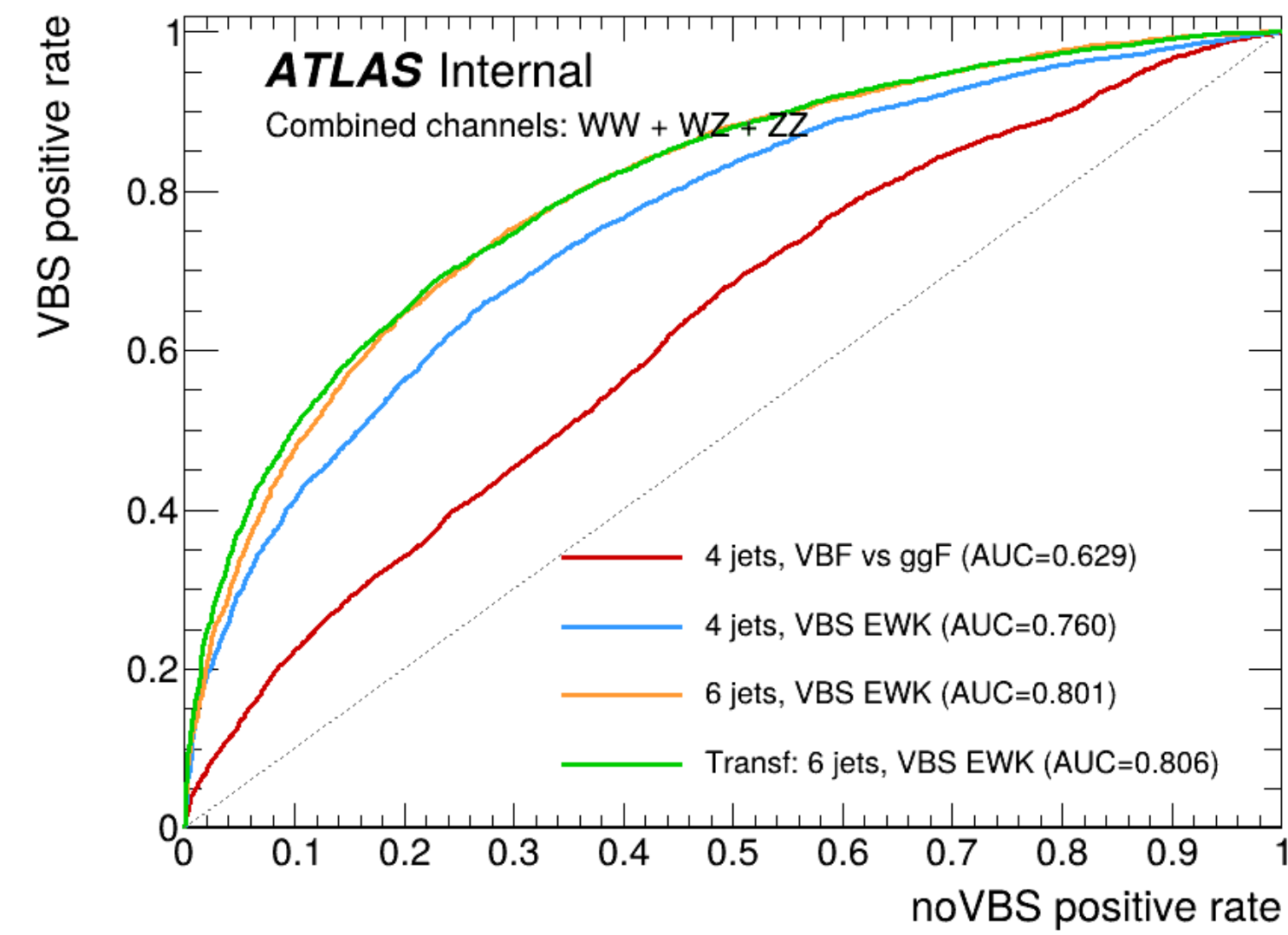
VBF Tagger

- Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM) cells
- Applied in previous analyses [5, 6]
- Designed to identify characteristic VBS/VBF topology
- Input: jet four-vectors (p_T , η , ϕ , E)
- Suited for short jet sequences

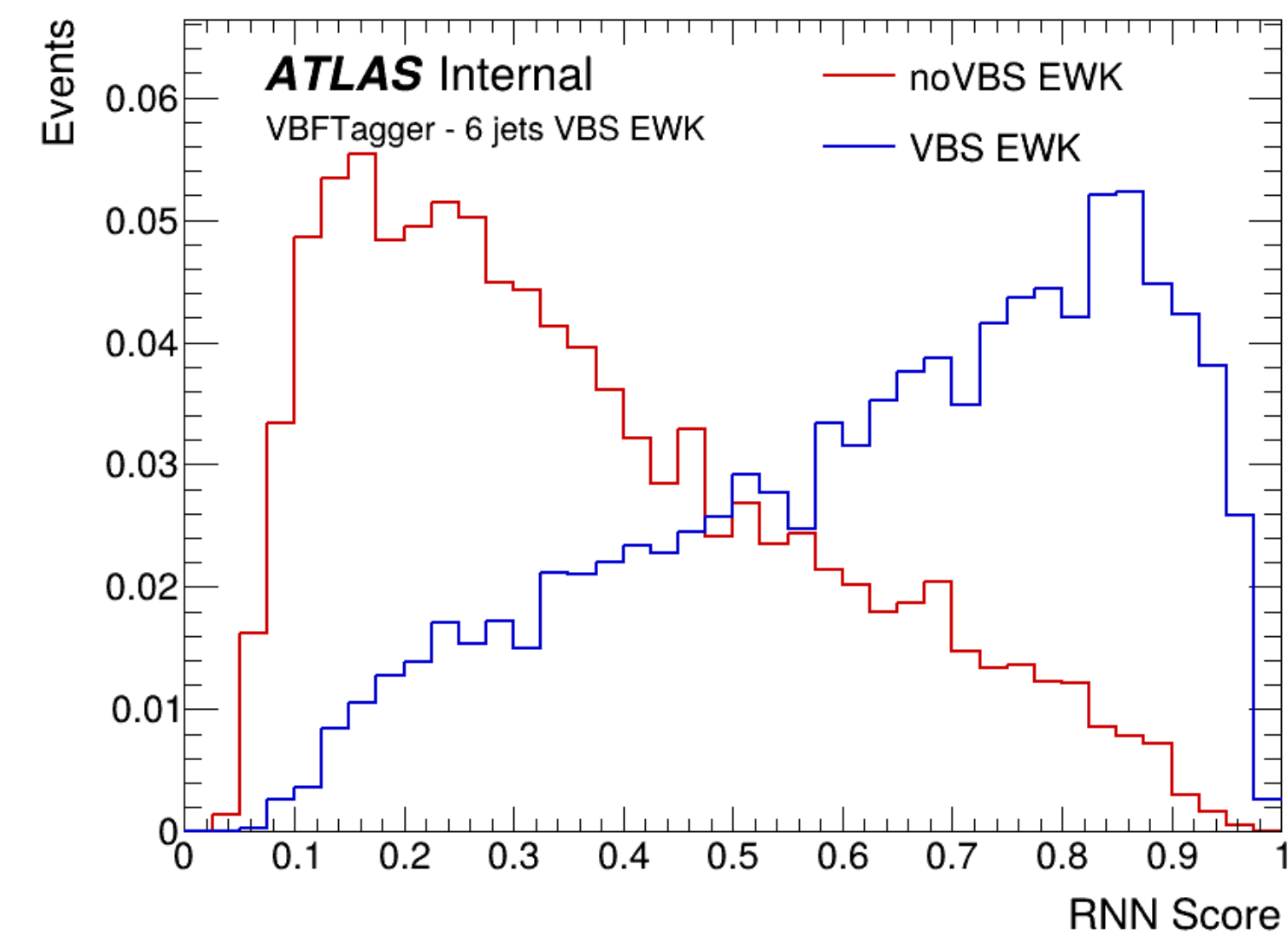
Version	Input	Training
v1	Up to 4 Tagging Jets	Resonance VBF vs ggF
v2	Up to 4 Tagging Jets	VBS vs noVBS (EWK)
v3	Up to 6 Jets (incl. Signal Jets)	VBS vs noVBS (EWK)

VBF-RNN MODEL PERFORMANCE

- Evaluated on separating **VBS** from **EWK noVBS**
- Inclusion of signal jets improves rejection of EWK noVBS background
- Transformer: marginal AUC gain, not significant at high VBS efficiency

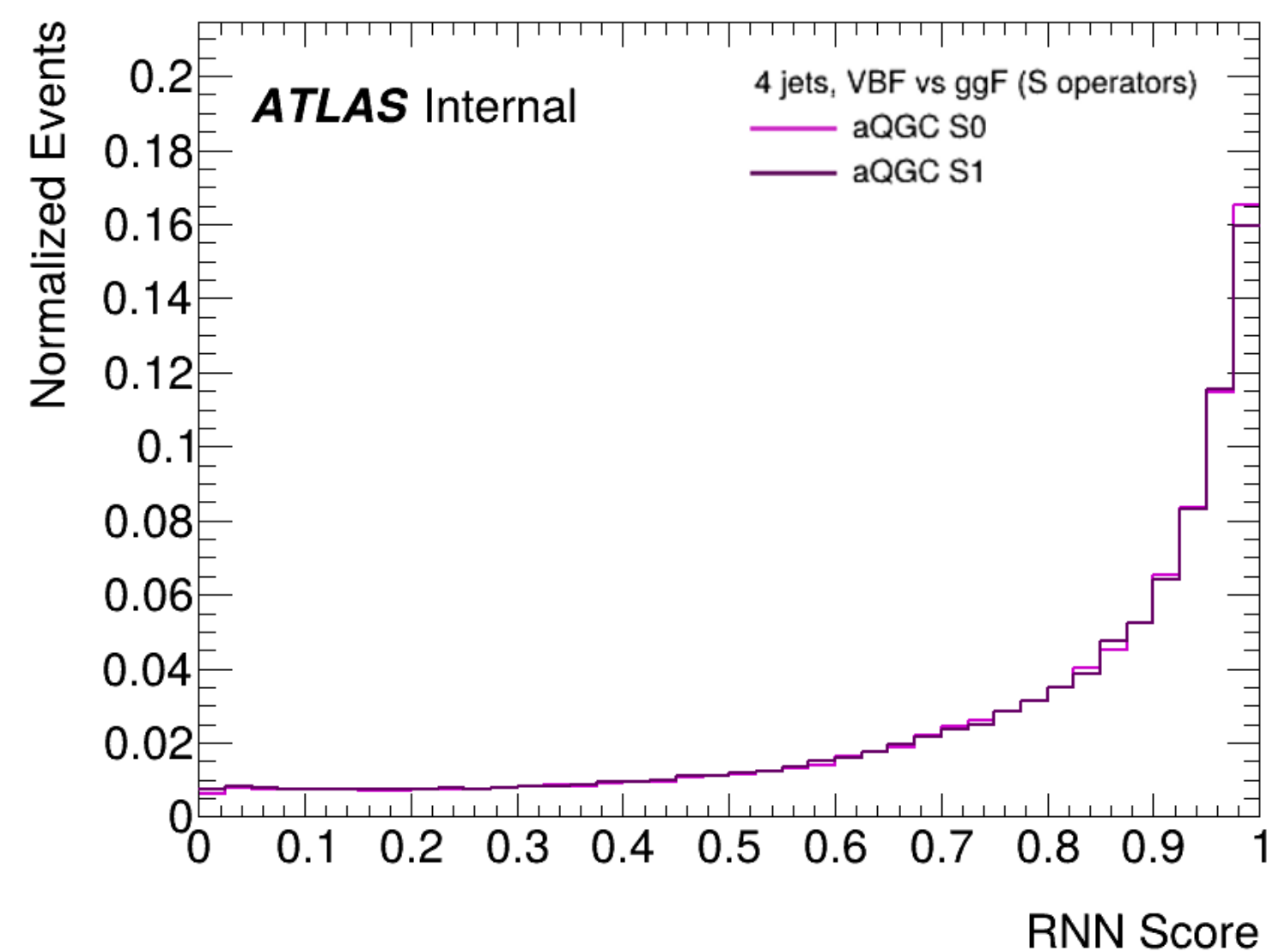


**Inclusion of Signal Jets
+5.4% AUC**

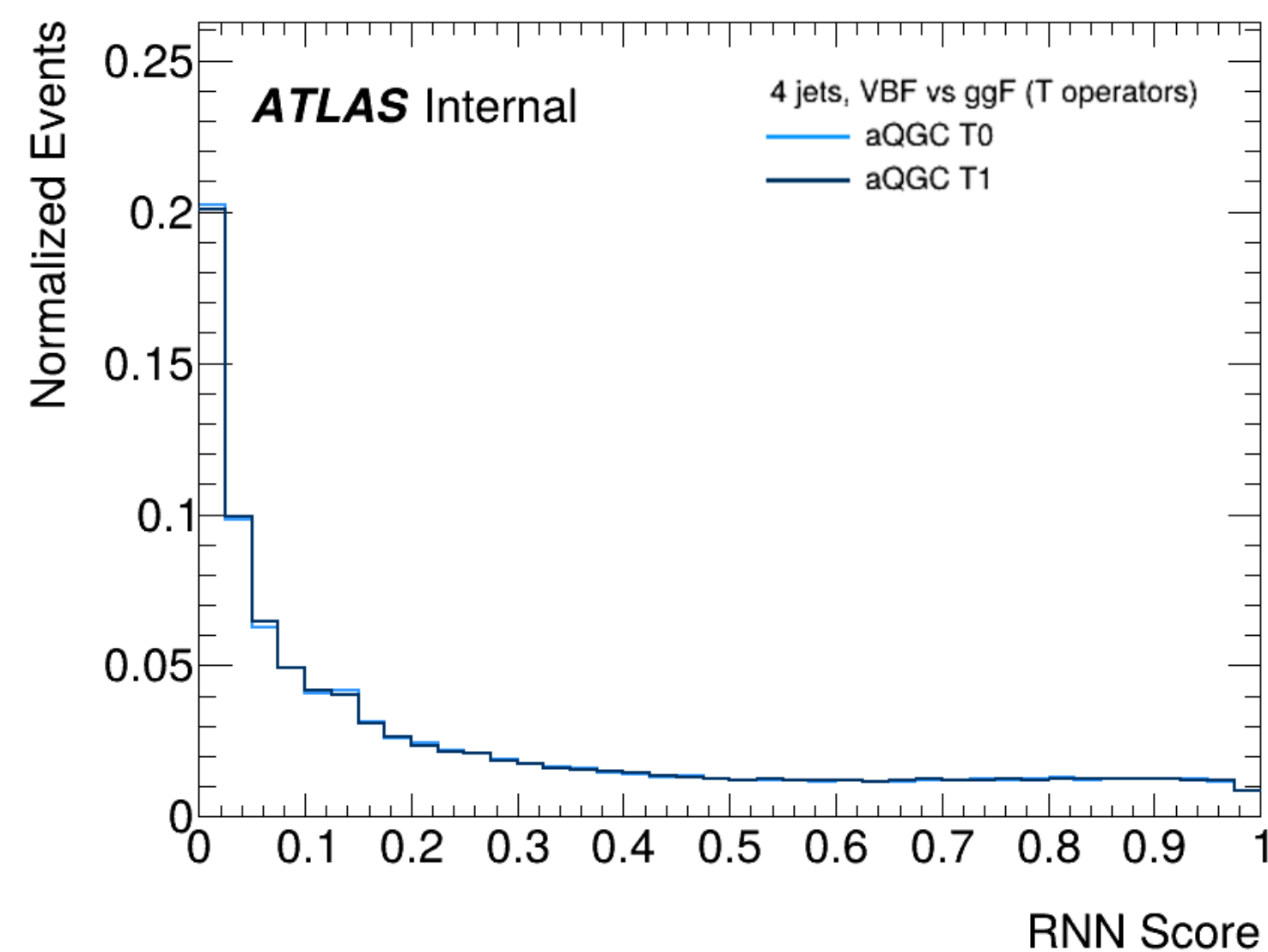


VBF-RNN AQGC RESPONSE

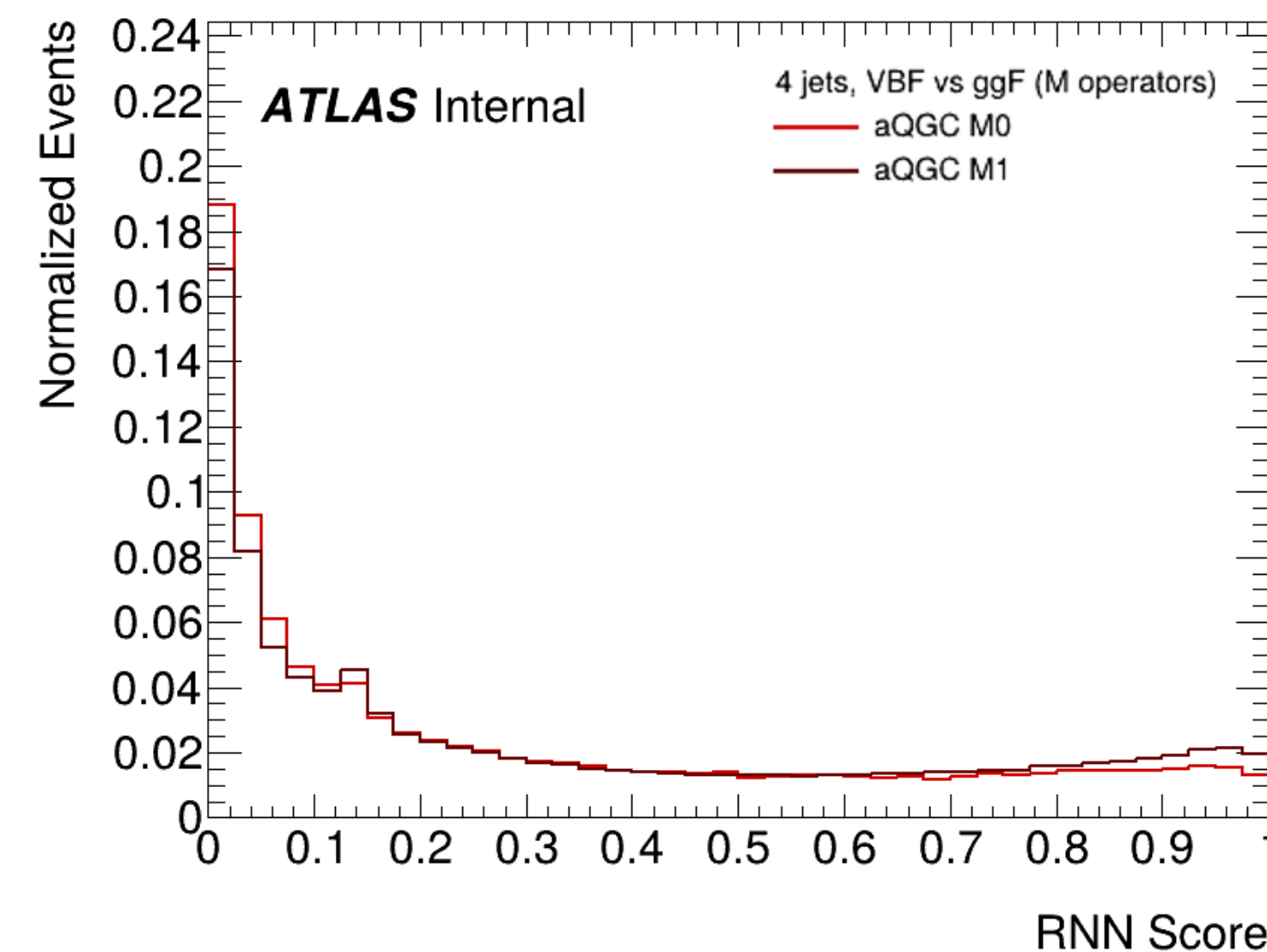
Version 1: 4 jets, Trained on: VBF vs ggF resonances



Scalar



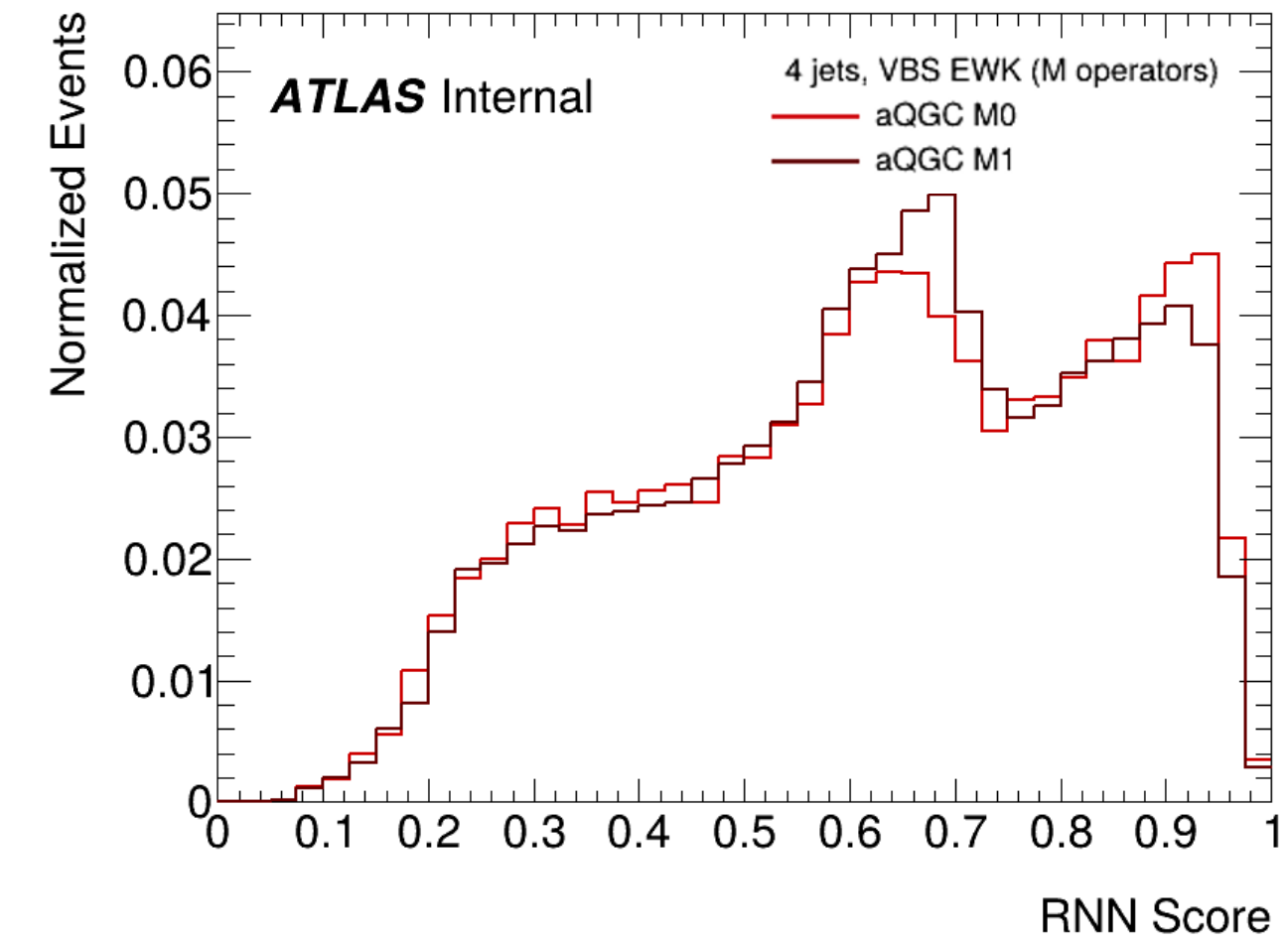
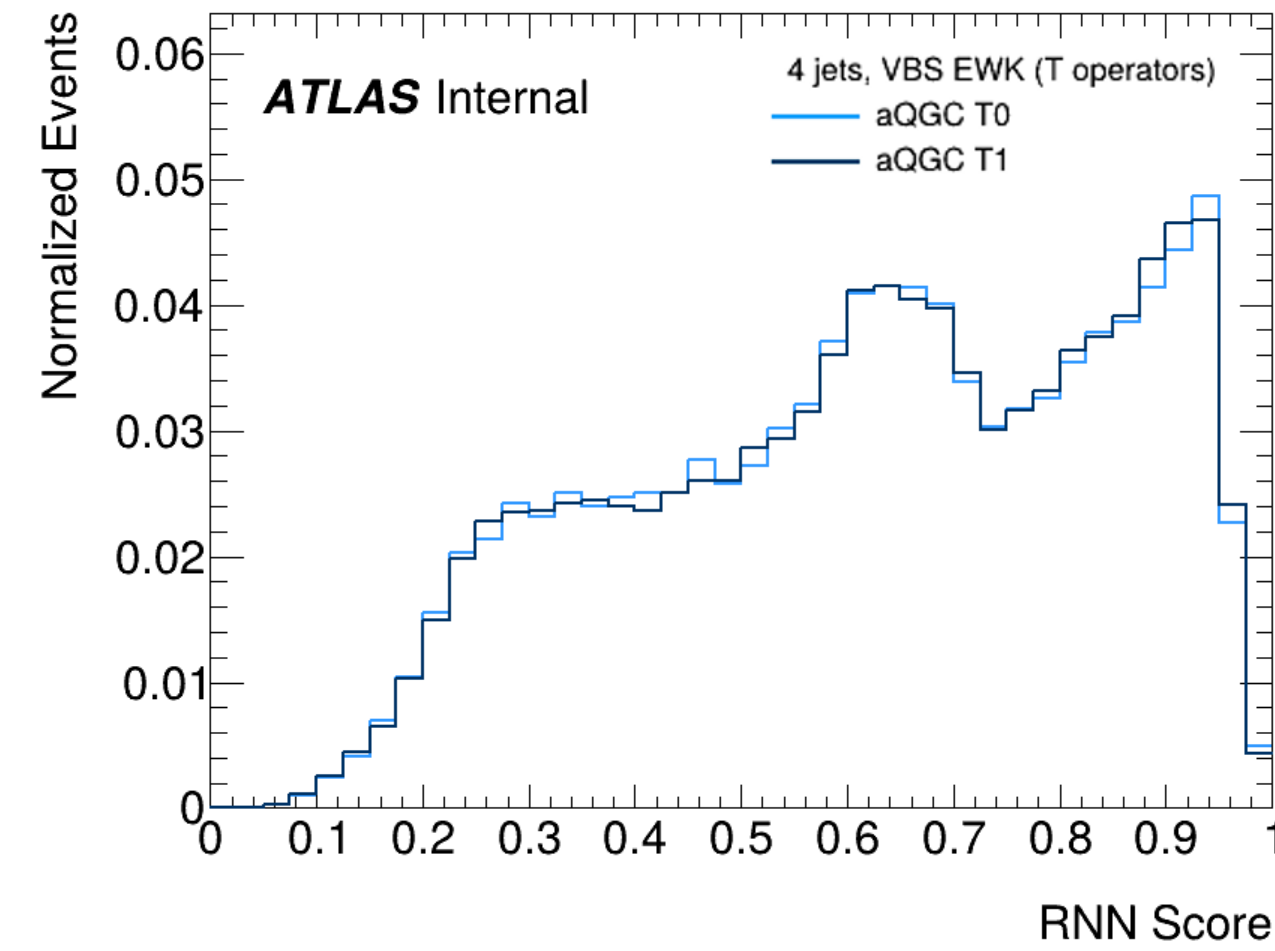
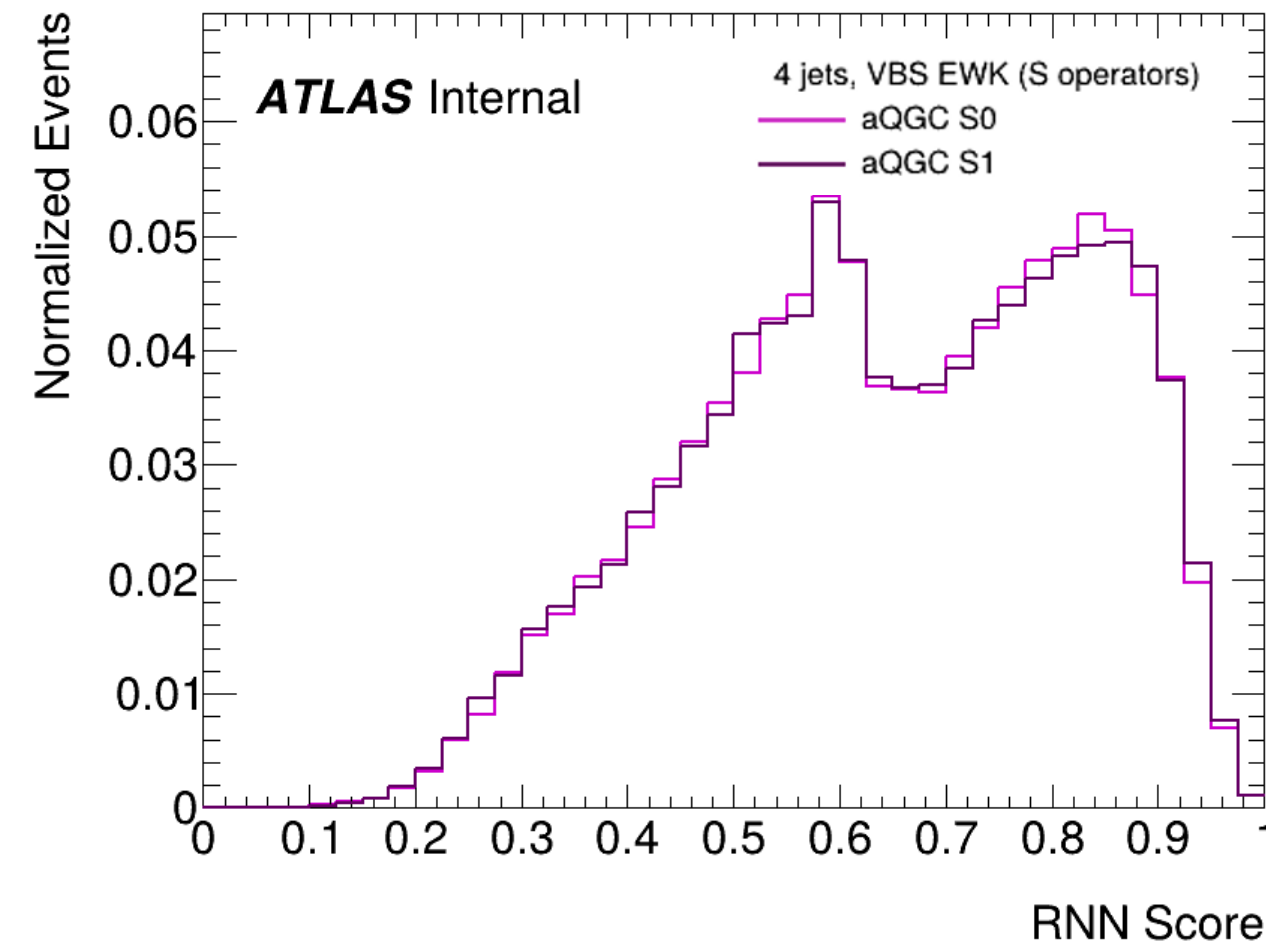
Tensor



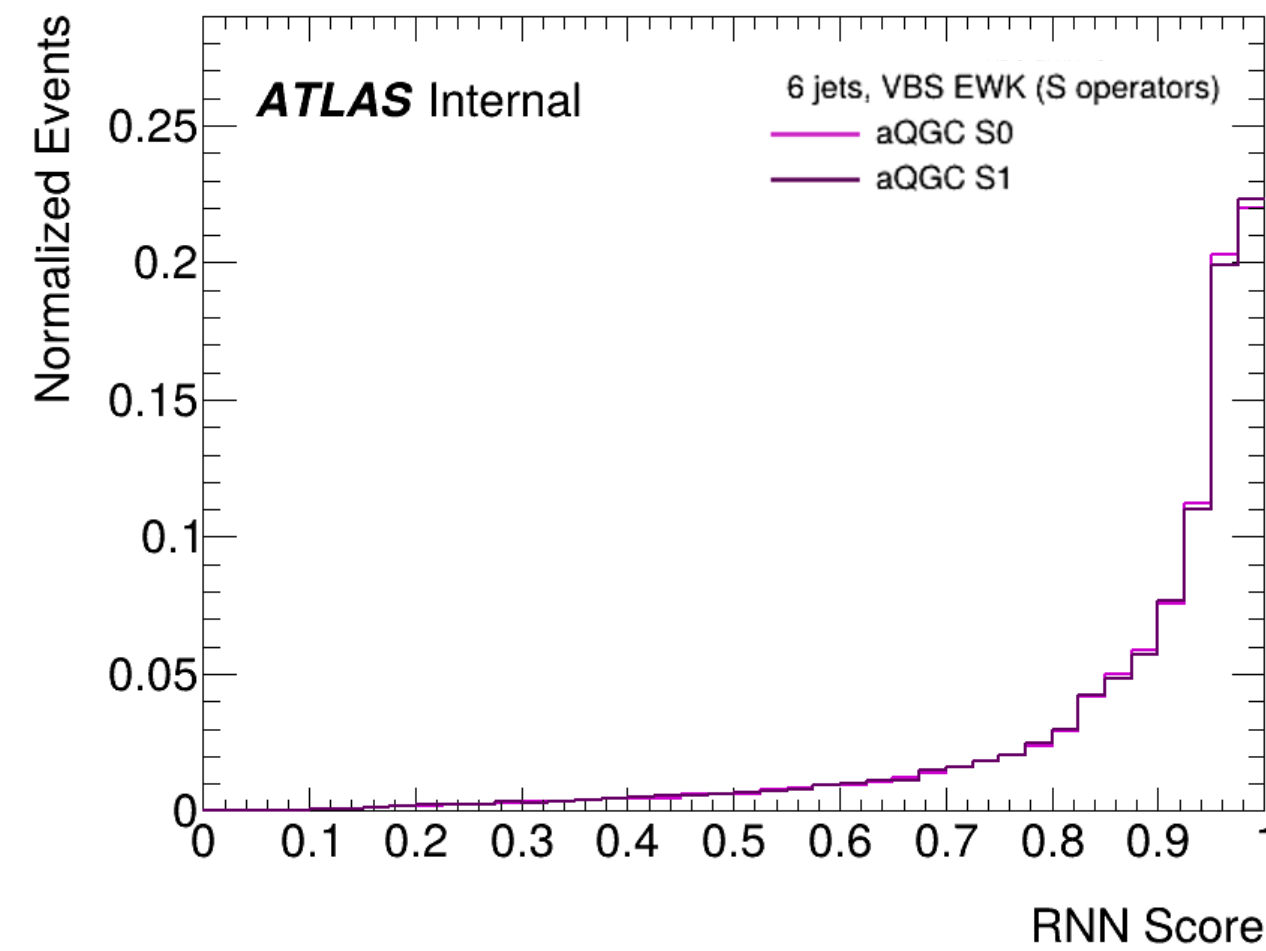
Mixed

VBF-RNN AQGC RESPONSE

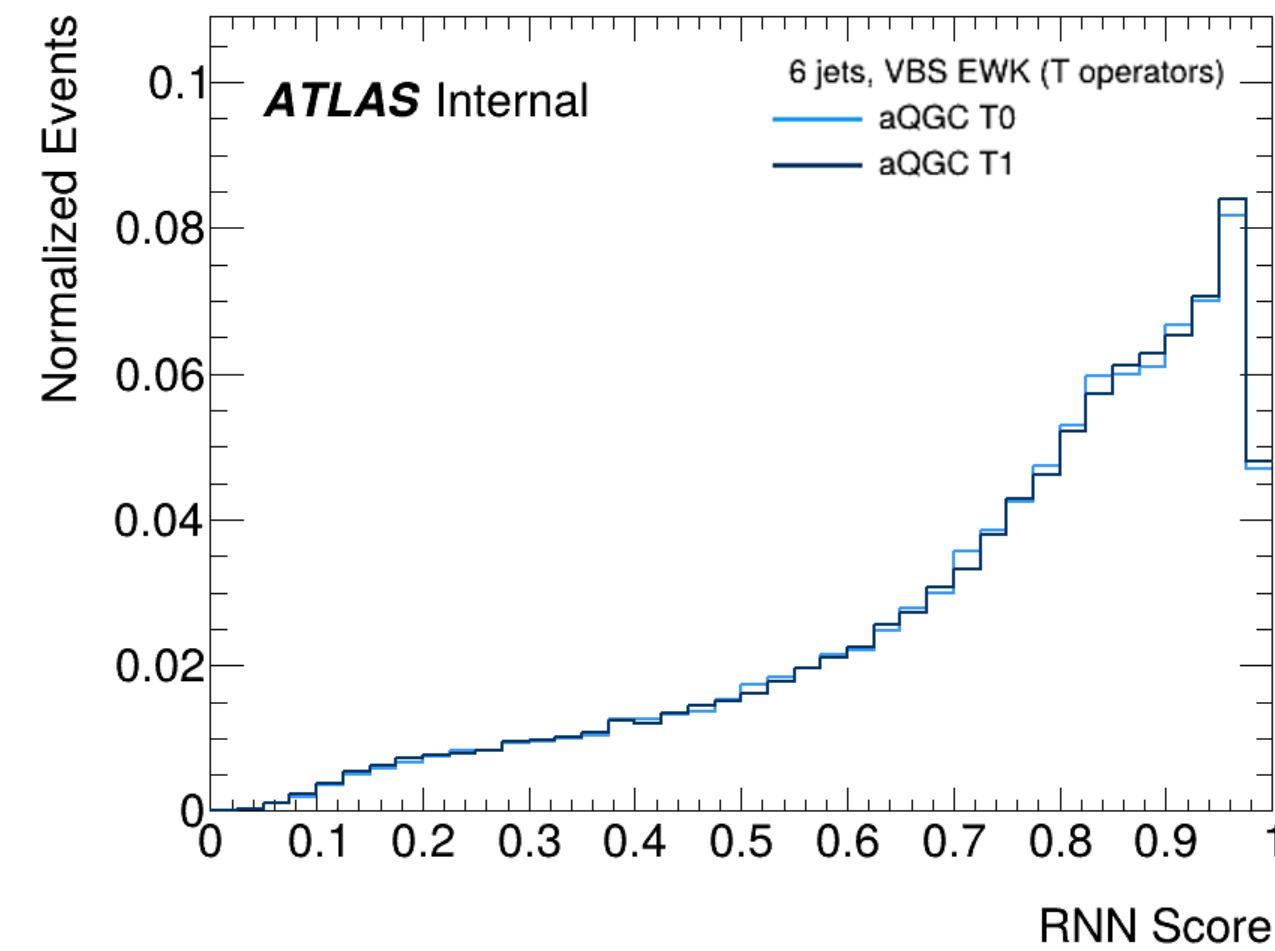
Version 2: 4 jets, Trained on: VBS vs noVBS EWK



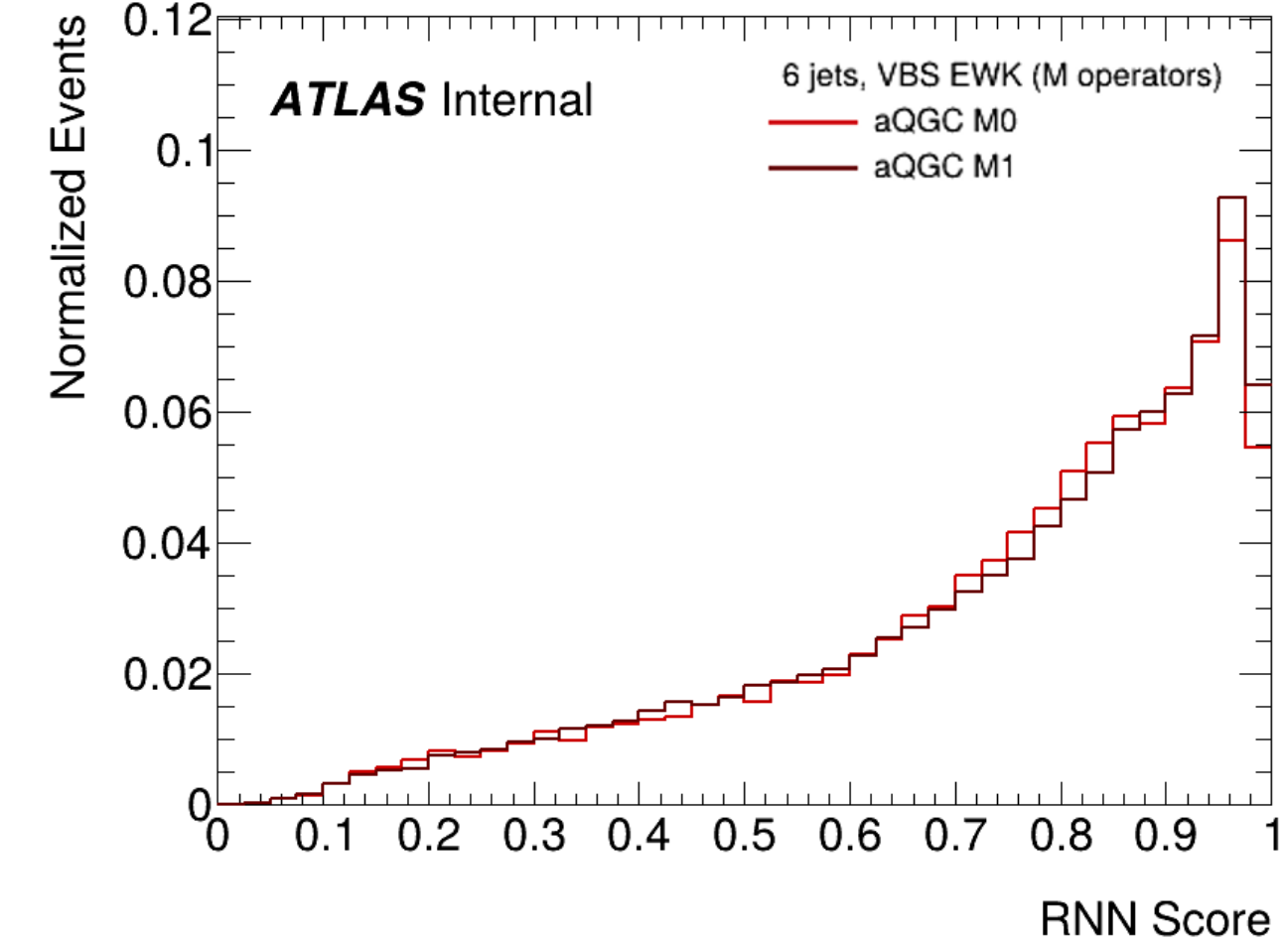
Version 3: 6 jets, Trained on: VBS vs noVBS EWK



Scalar



Tensor



Mixed

VBF-RNN

SUMMARY & FUTURE

What we achieved

- VBF-RNN successfully adapted for VBS all-had
- Identifies VBS topology for **all aQGC operator types**
- Important **input feature** for the ML discriminant

BACKUP: TRUTH INFO

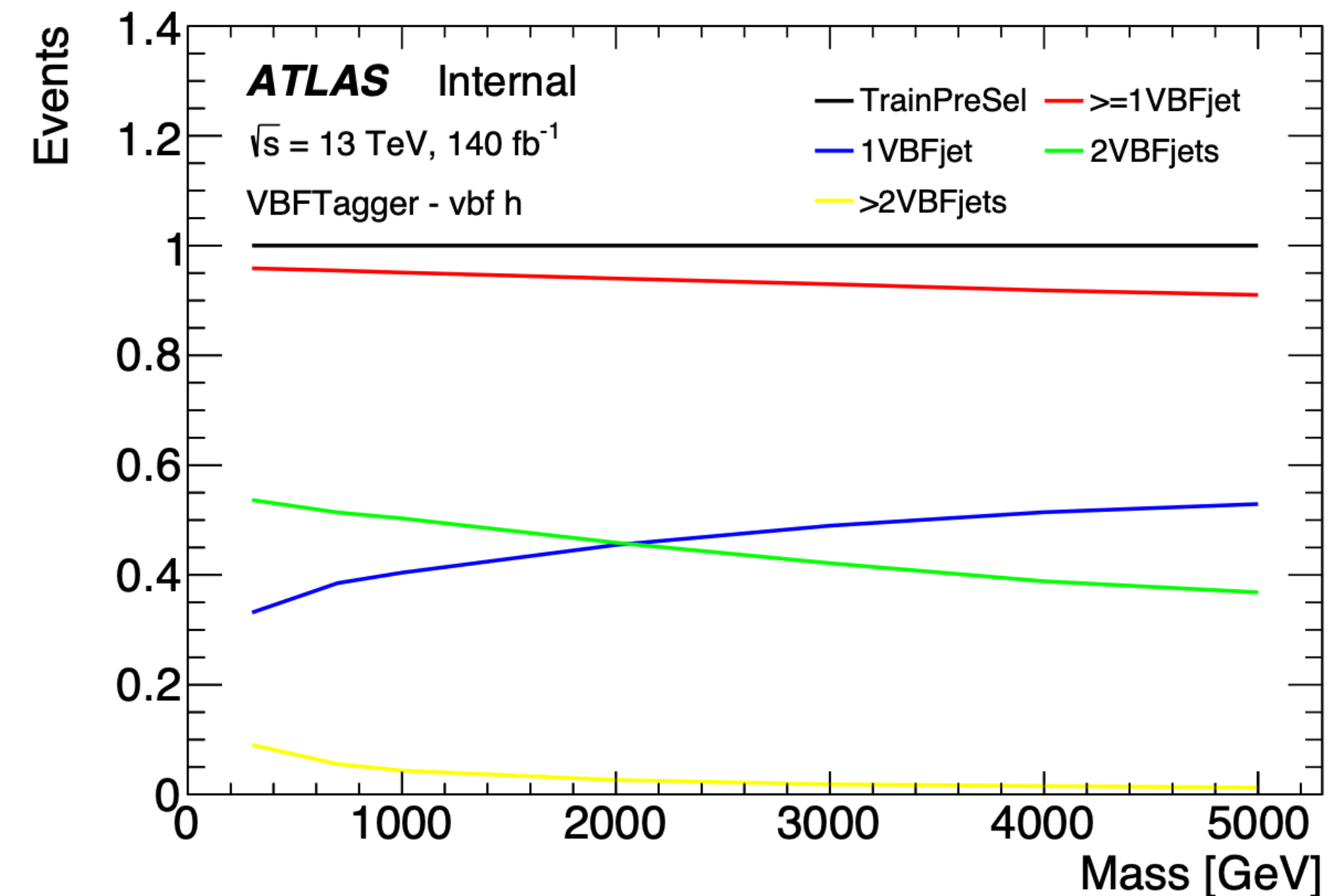
How we separate VBS from noVBS EWK from the EWK VV+jj samples

Boson matching: $DR(\text{jet}, \text{particle}) < 1$. (0.4) for large-R (small-R) jets

- isV: jet matched with a truth V
- isV1q: jet matched with a truth quark from V
- isV2q: jet matched with two truth quarks from V
- ish: jet matched with a truth V or h
- ish1b: jet matched with a b-quark from h
- ish2b: jet matched with two truth b-quarks from h

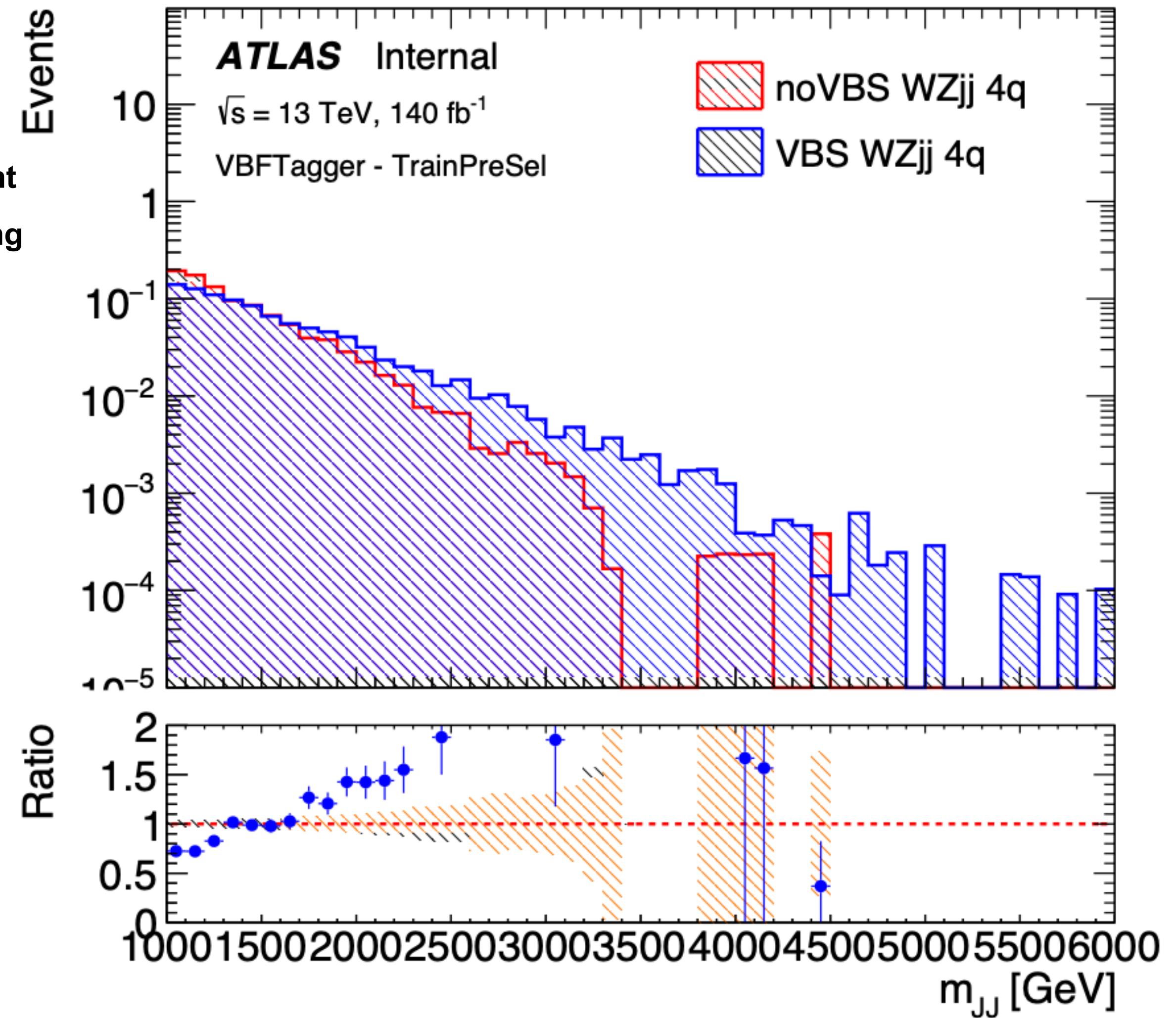
VBF matching: “by subtraction” strategy

- isVBF: jet that is NOT boson matched, not gluon initiated, not PileUp



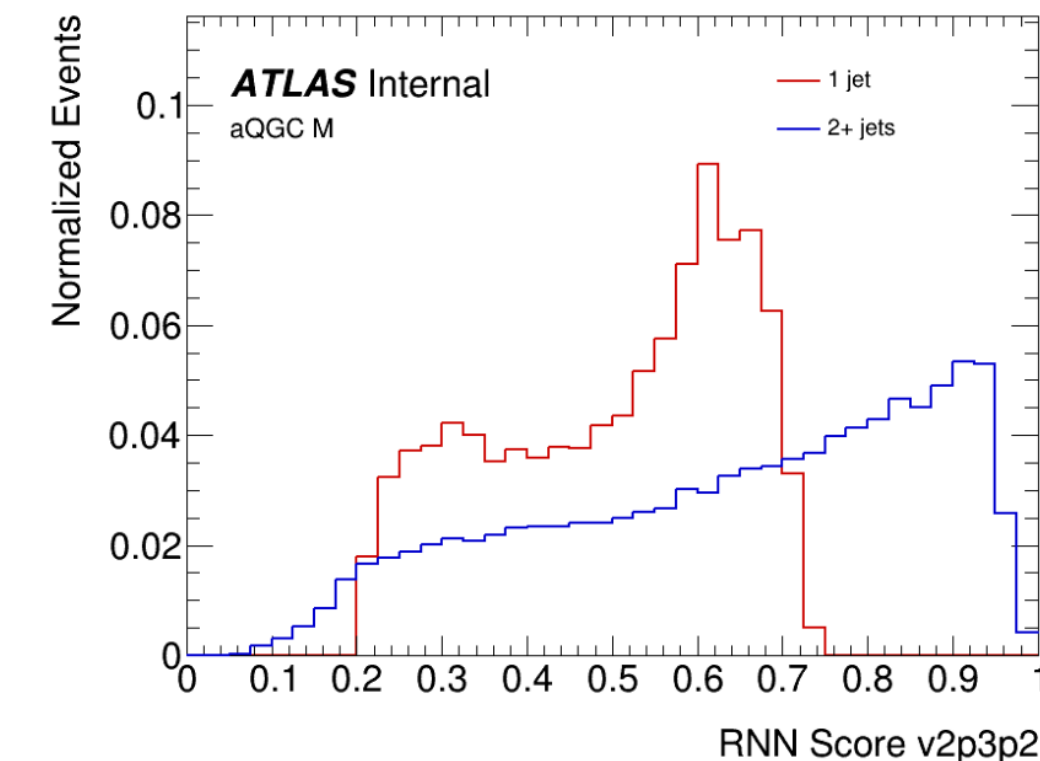
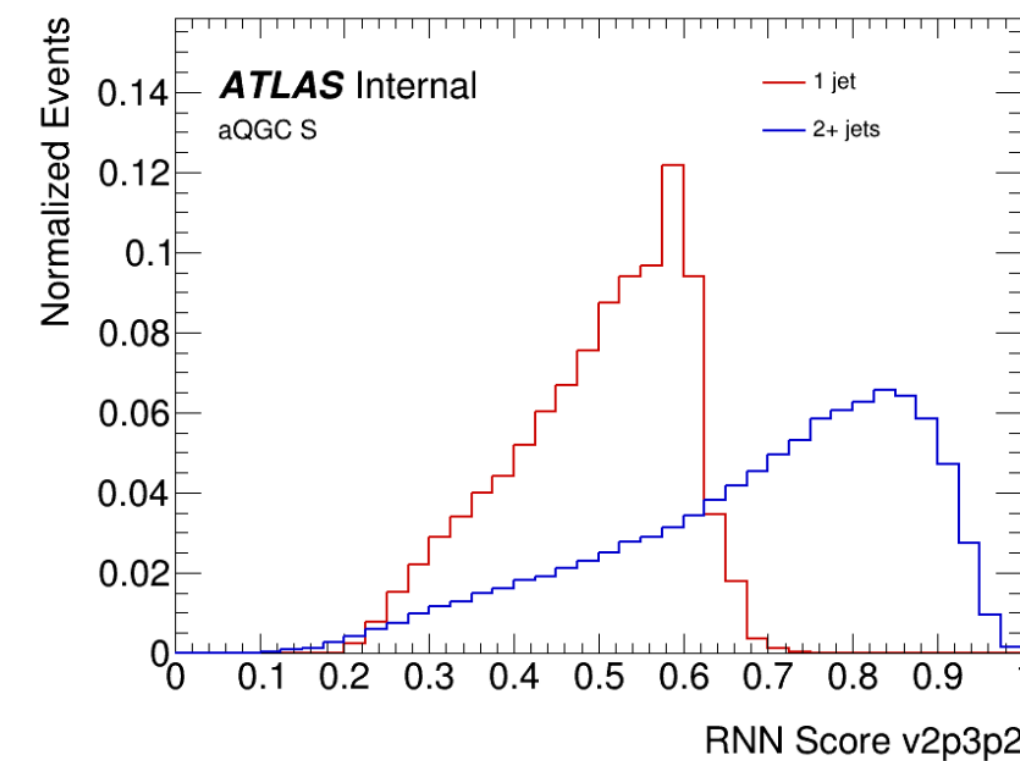
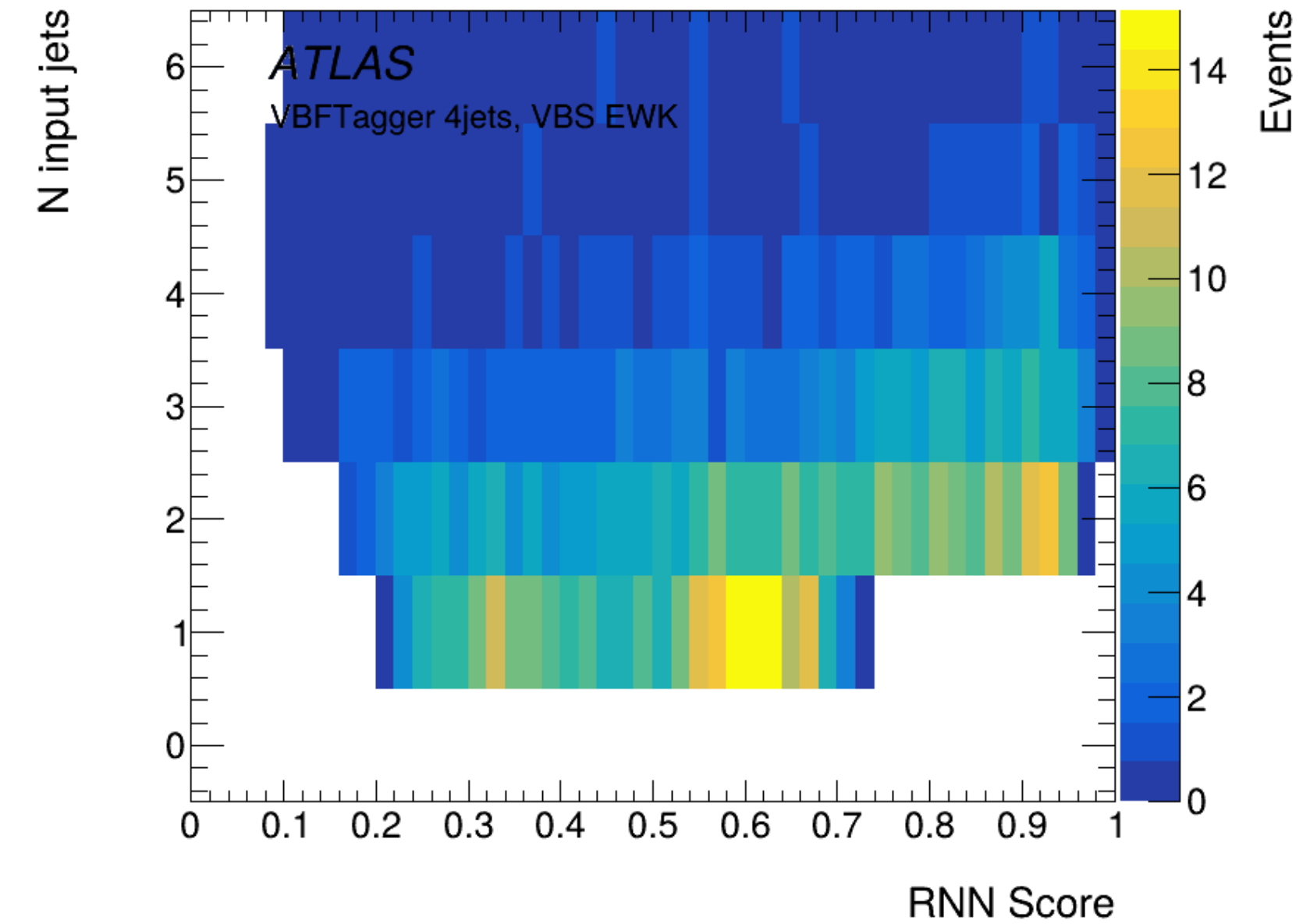
BACKUP: SIGNAL JET INFO

Still extra discriminating information in the signal jets dijet invariant mass, could including the signal jets help us improve discriminating power?



BACKUP: RNN JETS

- Number of jets that the RNN has as input, plotted against the given score of the event for the EWK, 4 jets model.
- Events with only one RNN Jet as input (due to detector response) fail to be assigned high RNN scores.
- This is what is causing the artifacts in the score distributions for the VBF-RNN v2



BACKUP: VBF-RNN CHANNELS

